

E4 Final Prototype Memo: Potato

Background

Professor Steinberg is a Political Science and Public Policy professor at Harvey Mudd College. He teaches and lectures from the Harvey Mudd Shanahan classrooms that have moveable chairs and tables for students along with a large and tall stationary desk for the professor. Professor Steinberg expressed that this setup prevents him from moving around freely, limiting his ability to naturally interact and maintain strong engagement with his students. The setup also prevents him from reading his notes at a comfortable height and angle. To address these problems, Prof. Steinberg is currently using a heavy and incredibly bulky mobile lectern. However, this lectern's outdated wood design distracts students.

Problem Statement

Professor Steinberg is looking for a way to hold his laptop and lecture notes that facilitates an open and engaging learning environment.

Prototype

Our final prototype is a mobile lectern that takes heavy inspiration from modern design that utilizes lattice patterns and structures in products as shown in Figure 1 & 2. Since utilizing this modern structure—the lattice—has been fundamental from our early designs, we generated the following prototype question: *Can the mobile lectern support different lecture note mediums while maintaining its modern design and aesthetic.*



Figure 1: Modern Lattice Structure Chair



Figure 2: Accordion Stretch

Final Functional Design Justification: Objectives and Constraints

Using the problem statement and background as guides, we developed a set of objectives, metrics, and constraints that would point us towards an ideation space suitable for the project. Using the piecewise comparison chart or PCC chart (Table 1), we ranked all objectives from the most to least importance as shown below.

1. Minimize weight
2. Maximize portability
3. Minimize set up time
4. Maximize durability
5. Maximize modern aesthetics
6. Maximize ecofriendliness
7. Minimize the number of parts of the solution
8. Minimize cost
9. Maximize the range of adjustable vertical heights

Check for logic error if there are ties in your PCC										
Total	Objectives	Minimize the number of parts of solution	Maximize modern aesthetics	Minimize set up time	Minimize cost	Maximize durability	Minimize weight	Maximize the range of adjustable vertical heights	Maximize portability	Maximize Ecofriendliness
2	Minimize the number of parts of solution		0	0	1	0	0	1	0	0
4	Maximize modern aesthetics	1		0	1	0	0	1	0	1
6	Minimize set up time	1	1		1	1	0	1	0	1
1	Minimize cost	0	0	0		0	0	1	0	0
5	Maximize durability	1	1	0	1		0	1	0	1
8	Minimize weight	1	1	1	1	1		1	1	1
0	Maximize the range of adjustable vertical heights	0	0	0	0	0	0		0	0
7	Maximize portability	1	1	1	1	1	0	1		1
3	Maximize Ecofriendliness	1	0	0	1	0	0	1	0	

Table 1: PCC Chart

We want to emphasize that portability functions as both a constraint and an objective for this project. Once a prototype satisfies the minimum portability constraint requirement, we will evaluate how well it maximizes portability.

To measure how well a prototype met each objective, we developed the following quantitative and qualitative metrics.

Metrics:

Minimize Weight	Quantitative: Mass in pounds	Score
	More than 5 pounds	0
	Between 4.6-5.0 pounds	1
	Between 4.1-4.5 pounds	2
	Between 3.6-4.0 pounds	3
	Between 3-3.5 pounds	4
	Less than 3 pounds	5

Maximize Portability	Qualitative: User's perceived portability	Score
	Does not meet constraint	-
	Slightly portable, very heavy and requires strength	1
	Portable but bulky, does not fit in a backpack	2
	Portable, can be hand held and carried	3
	Fits in regular sized backpack, carry with ease	4
	Easily transportable, fits in small laptop bag and can carry anywhere with ease	5

Minimize Setup Time	Quantitative: Time needed to set up	Score
	More than 5 seconds	0
	Between 4.5-5 seconds	1
	Between 4-4.4 seconds	2
	Between 3.5-3.9 seconds	3
	Between 3-3.4 seconds	4
	Less than 3 seconds	5

Maximize durability	Qualitative: User's perceived durability	Score
	Unable to stand independently	0
	Wobbly/shaky under no weight load	1
	Stable under no load, collapses under weight load of 0.9 pounds or less	2
	Stable until under moderate load of between 1-2 pounds. with noticeable shaking	3
	Sturdy, slightly noticeable movement under 2.1-2.9 pound load	4
	Very sturdy, no noticeable wobbliness even under a 3-5 pound load	5

Maximize modern aesthetics	Qualitative: User evaluation of design aesthetic	Score
	Design is unappealing, visually is out of place with the environment, gives the user an awkward appearance	0
	Design is somewhat modern, still out of place from environment and user appears unprofessional	1
	Design is lacking visually, disconnected from the environment, user presence is not affected	2
	Design aesthetic slightly matches the environment, disrupts the presence of the user	3
	Design aesthetic matches the environment and user has a professional presence	4
	Design is the ideal aesthetic which blends well with the environment and enhances the presence of user	5

Maximize ecofriendliness	Quantitative: % of sustainably sourced materials used in the lectern	Score
	24% or less	0
	Between 25 and 50%	1
	Between 51 and 75%	2
	Between 76 and 80%	3
	Between 81 and 90%	4
	Between 91 and 100%	5

Minimize the number of parts of the solution	Quantitative: # of parts of solution	Score
	More than 5 parts needed to be constructed	0
	5 parts for solution to be constructed	1
	4 parts for solution to be constructed	2
	3 parts needed to be constructed	3
	2 parts to construct the solution	4
	1 parts, one-piece design	5

Minimize cost	Quantitative: Total cost of design (USD) if on boundary, round up (e.g 50->51)	Score
	More than 50\$	0
	Between 45-50\$	1
	Between 35-45\$	2
	Between 25-35\$	3
	Between 15-25\$	4
	Less than 15\$	5

Maximize range of adjustable vertical height	Quantitative: Extended height range (inches)	Score
	0in (fixed)	0
	Less than 1inch	1
	Between 1-2 inches	2
	Between 2.1-3 inches	3
	Between 3.1-4inches	4
	More than 4.1 inches	5

Table 2

Using the metrics in table 2, we analyzed and scored our final prototype. Our final score was a 38/45, which we collectively deemed as satisfactory as most metrics received a score of 4 or a 5. The method of evaluation and justification for our scores are as follows:

Minimize Weight: We used a weighing scale to measure our prototype. We measured our prototype to weigh 2.6 pounds. According to our metrics, the prototyped scored a 5.

Maximize Portability: Each team member—with varying back pack sizes—attempted to put the mobile lectern into their respective backpacks. While only 2 team members with the larger backpacks were successful, we found this to be somewhat positive as other 2 team members had smaller sized backpacks at max capacity that prevented the solution being fit in it. Team members also carried the final prototype from Parsons to Sontag Hall and found that transporting the lectern requires minimal effort. As a result, we determined that our prototype scored a 4.

We would like to note that our client requested a sleeve to go with the prototype, as he dislikes the aesthetics of the mobile lectern when being transported. We have links to several sleeves already available on the market for our client to use. If the client is satisfied with this method of transportation and uses it, our prototype score will still remain a 4, as the sleeve is slightly bigger than a small laptop bag.

Minimize Setup Time: Our final prototype only requires decompression for a full assembly. When testing the set-up time for 5 different students in our E4 class, we found that the average time results in 1.6 seconds. As a result, this prototype scored a 5.

Maximize Durability: The team stacked four laptops—a total load of approximately 15 pounds—on top of the mobile lectern. Since the mobile lectern did not collapse or show any signs of instability, the prototype scored well above 5. We did not perform an FEA test to measure the stress the lectern can withstand as the test we have conducted was sufficient for the purpose of the lectern.

Maximize Modern Aesthetics: To assess the extent at which our lectern has a modern design, we interviewed our client and fellow E4 students. During the second client interview, he stated that he loved the mobile lectern's appearance when it's fully assembled because of its modern appearance. While he disliked the collapse view of the mobile lectern, we determined this comment to be negligible because it does not impact the engagement levels of the students. During our initial design review, we received data from the student audience that showed strong support for our design as its appearance was incredibly modern. As a result, our prototype scored a 5 on this objective.

Maximize Ecofriendliness: Our final prototype is primarily composed of tempered hardwood. Since tempered hardwood is recyclable, sustainably sourced and uses low amounts of emission during its manufacturing process, we determined this material to be highly sustainable. However, our prototype also uses small amounts of unsustainable materials in the form of magnets and contact paper. As a result, our prototype scores a 4.

Minimize the number of parts of solution: Our final prototype is a one-piece design with minor complementary details such as the popsicle wings that are not intended to be removed from the main assembly. Thus, our final prototype satisfies this objective and scores a 5.

Minimize Cost: Our prototype cost is \$24, scoring a 4. This is an expected result since our client had a low ranking for this objective as he understood that the tradeoff for maximizing modern aesthetics would be the cost.

Maximize range of adjustable vertical heights: Our final prototype has a fixed max height of 10 inches. Thus, this prototype scores a 0 in this objective. We decided not to focus on satisfying this objective as he stated he would like other professors to eventually use and manufacture their own mobile lectern. Given that it is simple to adjust the vertical heights of the design, our dismissal for this objective would be appropriate. We will like to note that for the lectern having a height of 10 inches was measured based off of our client's height.

The following are the constraints that we developed based on our initial client interview:

- Must be portable
- Must be assembled within 5 seconds
- Must be disassembled within 5 seconds.
- Must be useable for 3+ years
- Must weigh under 5 pounds

- Must be able to hold laptops and ipads while cables are plugged in
- Must be safe for the user to use
- Must be under \$50

The final prototype has met all of the constraints listed above making this prototype a feasible solution for our client. The prototype met each constraint using the following evaluations:

Must be portable: During our final client interview, our client stated that the weight in conjunction with the collapsible panels features makes this mobile lectern highly portable.

Must be assembled and disassembled within 5 seconds: Our prototype is able to be assembled and disabled within 5 seconds due to the collapsible panels feature. This feature allows the user to simply compress the outer corners of the lectern together, essentially flattening out the lectern when disabling the lectern. When assembling the mobile lectern, the user must remove any restraints that keep the lectern in its flat form to assemble the lectern. For verification, we asked 5 E4 students to disable the mobile lectern. With an average time of 1.6 seconds, it is clear that this constraint is satisfied by the prototype.

Must be usable for more than 3 years: Given that we cannot test to verify that the prototype will be usable for at least 3 years, we conduct brief research on the durability of the tempered hardwood—the primary material of the mobile lectern. Since tempered hardboards can last for at least 5 years when faced with weathering and other environmental conditions, the prototype meets the constraint.

Must weigh under 5 pounds: The prototype weighs only 2.3 pounds. As a result, this prototype satisfied the constraint.

Must be able to hold laptops and ipads while cables are plugged in: The prototype can hold any sort of technological device as well as paper notes without the device falling off. We verified this to be true by putting different electronic devices and sheets of paper onto the lectern. Since there were no obvious issues during these tests, we deemed this constraint to be satisfied. If the user decides to connect their electric device to a physical cable, the extendable wings will be used to provide support for the wings.

Must be safe for the user to use: To ensure this constraint was met, we had fellow E4 students operate the mobile lectern by placing notes, compressing, decompressing, and transporting the lectern. Given the positive feedback from the students, we concluded that the prototype poses no danger to the user when using the lectern.

Must be under \$50: The prototype costs \$26. As a result, this prototype satisfied the constraint.

Final Functional Design Justification: Material Choices

Our final prototype consists of the following six materials; tempered hardwood, gray contact paper, magnets, medium-sized popsicle sticks, duct tape, and rubber grip tape. Each of these materials were meticulously selected and served a highly dependent purpose as described below.

Tempered Hardwood: We selected tempered hardwood as the primary structural material because it best balances our objectives: minimize weight, maximize durability, maximize ecofriendliness, and minimize cost. In our earlier prototypes, we found that lighter materials like basswood and cardboard are less durable materials since they begin to crack after repeated folding. As a result, we decided to use tempered hardwood since it is a stronger material. Although tempered hardwood is slightly heavier, the trade off can be justified since the mobile lectern must be highly durable to meet our constraint of being usable for 3+ years. Additionally, tempered hardwood aligns with our sustainability goals quite well as the tempered hardwood is ecofriendly, more cost-effective than basswood, and readily available in hardware suppliers like Home Depot.

Gray Contact Paper: We selected gray contact paper to maximize modern aesthetics while minimizing cost. The contact paper enhances the professional appearance of the lectern and visually blends with the Shan classroom, helping the device complement the space without distracting from the user. Contact paper also aligns with our cost objective and meets our weight constraint. It is inexpensive, lightweight, and easy to apply, allowing us to improve aesthetics without pushing the prototype above the 5-pound constraint.

Magnets and Medium-Sized Popsicle Sticks: We used magnets and medium-sized popsicle sticks to create the expandable side wings. These materials enable the wings to extend and retract smoothly which meets our function that the lectern is able to support notebooks and laptops. This is because the wings allow the lectern to accommodate wider materials without requiring the user to constantly reposition their materials. Given that both materials are inexpensive and readily available, they minimize the overall cost of the mobile lectern.

Duct tape: Duct tape was used to secure the panels to prevent the interlocking pieces from slipping out of place during use. The lectern is assembled from 5 vertical panels: 2 base panels with fixed slots and 3 additional panels that slide down into those slots from the top. Since the three additional panels are only held in place by gravity, they can slip out of place when the mobile lectern is carried or adjusted. In placing the duct tape across the inner corner where 2 perpendicular panels meet, the sliding panel can be secured to a fixed panel. This prevents the 3 top panels from lifting or separating during use. The duct tape allows for the assembly to be stable while maintaining a lightweight, collapsible design and minimize cost because it is inexpensive and readily available.

Rubber Grip Tape: We selected rubber grip tape to maximize modern aesthetics while meeting the function to resist lateral forces. Because the grip tape is clear, it provides stability without visually cluttering the design. We placed grip tape to the top edges of the panel to keep the different mediums placed on the lectern from shifting during use. Additional grip tape was placed on the bottom of the lectern to reduce sliding on surfaces it rests on. Although the grip tape was an expensive material, it significantly improved stability and the overall quality of the design, making it a worthwhile choice.

Prototype Functionality

The mobile lectern's functions are as follows: supports notebooks and laptops, resists lateral forces, elevates notebook and laptop, and limits displacement during transportation.

Supports notebooks and laptops: Our first function is achieved through the beam-like support created through the vertical panels. These “beams” create a sturdy open surface that supports note-taking devices. Additionally, the lip acts as an antislip mechanism, ensuring that whatever item is placed at an angle does not slip. We also added expandable wings to the lectern, allowing for material with lower masses—such as loose leaf paper—to have extra support. To use the wings, the user must move and adjust the sticks to the upper magnet to activate the extra support.

Resists lateral forces: Our second function is achieved through the grip tape that is placed on the all mobile lectern surfaces that make contact with the table. This grip tape ensures that the mobile lectern remains in an expanded form when in use and resists lateral forces. We also added rubber grip tape to the top of the panels to prevent notebooks and laptops from slipping when the design is subject to lateral forces.

Elevates notebooks and laptops: Our third function is achieved through the height of the panels. When the mobile lectern is expanded, the height of the panels allows for any item sitting on the top surface to be elevated.

Limits displacement during transportation: Our fourth function is achieved through the collapsible features. In the mobile lectern being able to collapse laterally, it will allow the user to simply hold onto the center of the lectern and move around freely without worrying about the lectern disassembling unintentionally during transport. To further limit this, the addition of duct tape prevents the panels from falling out of their respective slits.

Evaluation Plan

Our team evaluated the final prototype of our product from a qualitative lens that examined the execution of the functions referenced in our final prototype question. Our final prototype question addresses feedback we received from the most recent client meeting which centered around his request for the lectern to accommodate loose leaf printer paper. This led us to more design interactions and verifying that newly built support features—extendable wings—are able to support paper.

Our evaluation plan is as follows:

1. Assemble the mobile lectern on a flat table
2. Place different types of lecture note mediums on slanted surface of the lectern
 - a. Recommended mediums to test
 - i. Computer
 - ii. iPad
 - iii. Loose leaf standard sized printer paper

- iv. Notebook (100 pages)
3. Interact with the note mediums by flipping through the pages, typing, writing, etc and note comfortability and stability of performing actions on lectern (see table 3)
4. Jostle the mobile lectern and record observations on how the lecture note medium behaves: does it fall off the lectern? Does it move around?

Criteria	Score	Description
Note Support	1	Lectern supports all lecture note mediums and interactions made with them (writing, typing, etc.) and moving or pushing around lectern does not cause them to fall off.
	3	Lectern supports all lecture note mediums but it can be difficult to interact with certain types or move things around. Lectern supports most note mediums even when the lectern system moves around.
	5	All interactions with notes do not cause notes to fall off lectern or knock over lectern as a whole, Shaking the table does not cause things to fall off the lectern
Lectern Weight Stability	1	Withstands no weight, barely a piece of paper
	3	Withstands 5-10 pounds of force
	5	Withstands 15+ pounds of force

Table 3

Evaluation Results

This final prototype includes attachments that accommodate further support for paper notes as it provides enough support to hold two pieces of standard sized printer paper—side to side—without overlap. The added wing extension gives extra surface area for paper without sacrificing the small size of the current compressed form. The trials consisted of using the following note mediums and corresponding notes:

- iPad: This is well supported by the lectern and there are no signs of iPad slippage. As a result, the interactions with an iPad—such as writing on the ipad—can be performed with ease.
- Laptop: The elevation is good for people standing and viewing the screen while the angled surface is ergonomic for typing. The lectern supports this interaction and can hold 4+ computers on its surface.
- Notebook: The notebook is well supported by the mobile lectern and shows no signs of slippage. As a result, the interactions with an notebook—such as writing on the ipad—can be performed with ease.
- Printer Paper: This medium is well supported on the lectern. There are no obstacles in flipping between and moving pages around on the lectern as the wings offer extra surface area support for

multiple sheets of paper to rest on its surface (30+). This stops the loose leaf pages from hanging precariously on the edge or falling off.

During the note support test, it was observed that the lectern was able to provide support for different types of lecture note mediums. As a result, this test received a score of 5/5. As mentioned previously, the mobile lectern is able to withstand at least 15 pounds, a weight that will most likely not be placed on the lectern. To further examine how much more weight the lectern can withstand, we added 25 pounds of miscellaneous items onto the lectern. Given that our lectern remained stable, this test received a score of 5/5.

These trials provided us with key insights from a user perspective that bolstered confidence in our final prototype. This lectern incorporates side wings attached to the horizontal pieces to give support to printer paper notes without sacrificing the size of the lectern and causing a much longer shape when compressed. Our recent meeting with him before creating our final prototype cemented this and provided our client with a product that can replace the large stationary lecterns blocked off in the corner of the classroom or the makeshift lectern he currently builds with miscellaneous textbooks at the beginning of each class. Our mobile lectern allows the client to engage with the audience better, opting as a less “barrier-like” alternative to those large immobile chest height lecterns that isolate him much more from his audience and that he addressed as a major pain point during our client meeting.

However, there is always opportunity for further optimization. As mentioned previously our client dislikes the compressed view of our mobile lectern as it does not evoke the same air of modernity and simplicity he values. Additionally, the mechanics of our current lectern’s compression property requires a consistent force to remain closed. This was originally a design choice to ensure that the lectern does not slowly close on the lecturer’s notes as the class progresses and makes it less prone to expanding over time. While our current solution incorporates a sleeve or just placing it in a bag to ensure its compression, our team would love to explore other options such as a drawstring cord or cinch cord. This could be attached through all central panels of the lectern and keep the lectern in its closed position without the need for a secondary detached component like a sleeve. With the small size of a drawstring cord and the structure of the walls, we could attach it so that it would be invisible from the outside and therefore not clash with the simplistic design of our lectern.